Vo Vinh Truong (Vinton - 武永長)

Biomedical Engineer | Computer Vision for Medical Imaging
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SUMMARY

Biomedical engineer with hospital X-ray (DR) field experience, building AI for pediatric fracture detection.

EDUCATION

National Taipei University, Taiwan 2026 – present
Master's Program in Artificial Intelligence Application and Management, NVIDIA INTENSE program

National Taiwan University of Science and Technology, Taiwan 2024 – 2025
Graduate coursework in Electronic and Computer Engineering

HCMC University of Technology and Education, Vietnam 2019 – 2023
B.Eng. in Biomedical Engineering, with Distinction

RESEARCH & PROJECTS

Master's Thesis: Pediatric Wrist Fracture Detection with YOLO26 and Mamba 2026 – present
- Fine-tuned YOLO26 (n to x) on 20,327 pediatric wrist X-rays; baseline mAP50 up to 0.933.
- Built a patient-level data split and multi-seed input-size experiments.
- Integrating a Mamba state-space block into the YOLO26 backbone.

Autonomous medicine-delivery robot for clinics (team of two) 2023
- Designed and implemented a Deep Q-Network route planner and the 2D simulation used to train it.
- Simulated the circuit in Proteus; programmed an Arduino Mega with FreeRTOS and sensor modules.

WORK EXPERIENCE

Field Engineer, Hai Nam Medical Service Trading Company, Vietnam Sep 2023 – Sep 2024
- Installed a DR digital X-ray system and PZDR imaging software at a district hospital.
- Performed preventive maintenance, troubleshooting and power-board repair on medical equipment.
- Trained hospital technicians and served as their link to the R&D team on technical issues.

Vietnamese Speech Data Contributor (part-time), Everbot Sep 2026 – present
- Record and transcribe Vietnamese speech for ASR and TTS fine-tuning, covering daily and healthcare topics.
- Revise audio, transcripts and metadata based on biweekly cloud ASR review.

Part-time Warehouse Assistant, Nippon Express Taiwan Jul 2026 – present
- Order picking, packing and put-away, 2 days per week.

Teaching Assistant, HCMC University of Technology and Education 2022 – 2023
- Supported Embedded Programming and Biomedical Image Processing courses.

SKILLS

- **ML & Computer Vision:** PyTorch, object detection (YOLO26), state-space models (Mamba), explainable AI (EigenCAM), reinforcement learning (DQN)
- **Programming & Data:** Python, C (Arduino, FreeRTOS), NumPy, Pandas, Matplotlib
- **Infrastructure:** Linux GPU servers (SSH, CUDA, Anaconda)
- **Hardware & Medical Devices:** DR X-ray systems, medical equipment maintenance, Proteus

HONORS & AWARDS

- **INTENSE Program Scholarship (Ministry of Education; NVIDIA)** 2026
- **Full Scholarship, National Taiwan University of Science and Technology** 2024
- **Top 5 GPA Graduation Scholarship, Biomedical Engineering, HCMUTE** 2023
- **Academic Excellence Scholarship, HCMUTE** 2023

LANGUAGES

Vietnamese: native · **English:** TOEIC 800 · **Chinese (Traditional):** TOCFL Level 3 (進階級)